Coding Discussion: Pruned Exhaustive (Branch and Bound) Search

Design and Analysis of Algorithms

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Traveling Salesman

- Input is N cities.
  - Here, we'll use points in the 2D plane so we can more easily visualize answers.
- Goal: find the shortest "tour" visiting each of them exactly once.
- NP-hard problem, so solving large instances optimally expected to be out of reach.

Traveling Salesman : Greedy

- Start at city 0 (arbitrary; could have picked a different starting point or tried many).
- Then move to the closest point.
- Then move to the closest remaining point.
- Etc.

- Fast ($O(n^2)$)
- May generate reasonably good solutions, which could then be further refined.

Traveling Salesman : Recursive Exhaustive Search

- Start at city 0 (an arbitrary choice).
- Loop through all possible points as the second point in the tour.
- For each of these, recursively loop through all possible points as the third point, etc.
- Remember best solution as we go.

- Optimal but slow!! Running time O(N!)
- Sequential decision / tree outlook.

Pruned Recursive Exhaustive (Branch and Bound) Search

- Do the recursive search just as before.
- Keep track of best solution found so far (the "incumbent" solution).
- If we ever build a partial solution that has no hope of being extended to a full solution better than the incumbent solution, "prune" search (i.e., back up and try other possibilities)
- When branching, try options in greedy order.
- (Sometimes relevant, but not so much TSP)… if we ever build an infeasible solution, prune search as well.